

Philip Booth

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EDUCATION

Colby College

Majors: Data Science & Classics (GPA: 3.85)

Waterville, ME

Graduated: May 2026

- Relevant coursework: Interpretable AI & Machine Learning, AI Ethics Independent Study, Multimodal Interaction, Data Visualization, Data Structures & Algorithms, Linear Algebra

SKILLS

Languages: Python, SQL, R, Java

Data Engineering: Data pipelines, ETL workflows, data ingestion, distributed data processing, Apache Spark

Tools: Git, Docker, Bash, GCP (BigQuery, Cloud Functions), AWS (Lambda), PostgreSQL, Snowflake

Libraries: pandas, NumPy, scikit-learn, Matplotlib, OpenCV, PyTorch, ChromaDB

DIRECT EXPERIENCE

AI Engineer Intern

May-Aug 2025

Props (Performance Content Marketing Platform), New York, NY

- Designed a Python ETL pipeline that ingested and processed 6,000+ video and image assets, generated structured metadata via automated classification, and loaded results into BigQuery to power the client analytics dashboard.
- Automated a manual asset deduplication workflow, detecting and filtering near-duplicate media assets before they entered the pipeline and improving reliability at scale.
- Reached ~94% classification accuracy by designing and running LLM evaluations across cost and performance dimensions using OpenRouter, iterating on prompt design.

Data Science Intern

May-Jun 2024

Penguin Random House, New York, NY

- Queried large Snowflake datasets with SQL to analyze national book sales trends and forecast the impact of regional book bans, presenting findings to the legal team to inform litigation strategy.
- Improved marketing data reliability by cross-referencing scraped AI-generated endorsements against structured source parsing to detect and document hallucinations, preventing inaccurate content from entering downstream analysis.

Sales & Operations

Jul-Aug 2024, Jun-Aug 2023

A2 Aviation, New York, NY

- Negotiated acquisition of airplane parts through bid/ask processes, analyzing historical sales data to optimize pricing and condition across tight-deadline transactions.
- Developed resale strategies and coordinated purchase orders with major carriers including Delta and United, generating over \$250,000 in revenue.

PROJECTS

ASA DataFest 2026, Best Insight Award (R, Python, Plotly)

- Won Best Insight at ASA DataFest 2026 by building a Random Forest model with SHAP values to analyze patient journey data, revealing critical gaps in behavioral health access to primary care.

ShakespeareGPT (Python, ChromaDB, OpenAI API) | [Live Demo](#)

- Built an end-to-end RAG system over Shakespeare's complete works, scraping and embedding texts into ChromaDB with optimized chunking and metadata indexing.
- Implemented hybrid sparse/dense embeddings to improve retrieval precision across semantically similar passages.

Additional: Ancient Greek OCR • MCP for Latin Dictionary • Marathon Data Viz & EDA • NFL Combine Data Analysis

LEADERSHIP

Captain, Club Soccer

2024-2026

- Led a competitive collegiate team to an undefeated season and regional tournament qualification by managing facilities, budgets, and roster decisions.

President, Classics Club

2023-2026

- Expanded Classics department engagement by organizing faculty talks, film screenings, and trivia nights.